Abstract

The aim of this sub-project is to develop an image processing component to support low-cost automated greenhouses. The image processing application must enable the greenhouse system to establish various aspects about the plants from the images taken. These aspects include plant health, plant growth and plant disease identification from the images of different parts of the plants in the greenhouse.

What is the purpose of this section?

- Context and importance of your research.
- Outline of literatur review

Why choose image processing techniques?

- Former methods cost too much time and were expensive + destructive [17]
- Focus on image analysis component [10]
- Review plant phenotyping concepts in relation to image processing and plant breeding, discussing various measurable traits

1 Plant Phenotyping

Plant breeding is defined as the process of inducing desirable inheritable traits in plants through controlled human action [26], which is important in solving the problem of providing an adequate global food supply. The rapid growth of the human population has created a strain on international food security. It is estimated that crop production must increase by 70% by 2050 [7] to meet this demand. Additionally, abiotic and biotic stressors limit cultivar biomass and yield [12]. Therefore, plant breeders need effective techniques to produce germplasm for cultivars that maximize yield and biomass, in various adverse environmental conditions. In the last few decades, image processing techniques in plant phenotyping have been an effective tool for plant breeding [3].

1.1 What Is Plant Phenotyping?

Plant phenotyping is the evaluation of phenotypes in plants [11], where phenotypes are primitive or complex traits of an organism that can be observed or quantified and evaluated numerically [15]. Phenotypes can alternatively be seen as a combination of all morphological, physiological, chemical, and behavioural aspects that represent the individual plant [13, 23, 20].

1.2 Plant Phenotyping As Method For Improving Plant Traits

Quantitative data produced by plant phenotyping is most useful in analyzing plant-environment interactions in plant breeding [11] for the development of germplasm with improved plant traits [20]. The data produced by phenotyping is used as a proxy for assessing a plant's performance in relation to its environment. For example, stressors or disturbances of plants are induced by environmental conditions that organisms are placed in, and these stressors induce symptoms which can be detected and evaluated using phenotyping. This indicates phenotypical traits of a plant have a complex dynamic feedback loop with their environment [36]. Therefore phenotypical traits serve as proxies in determining whether certain genotypes have favourable traits in particular environmental conditions.

1.3 Image-Obtainable Phenotypical Traits As Indicators Of Plant Health, Growth, And Stress

A phenotypical trait is classifiable as morphological, physiological, or temporal. Additionally, the literature indicates morphological and physiological phenotypical traits are classifiable as holistic-based or component-based [8]. Holistic properties consider the plant as a whole, whereas component-based features look at organs like the organism's fruit, flower, leaves, stem, etc.

Leaf-derived traits are one of the most commonly used plant status indicators because leaves rapidly respond to environmental stimuli and robustly indicate various factors that determine plant health, growth and stress. [40].Traits of leaves can monitor and predict a host of plant health, growth and stress factors. For example, leaf area predicts plant productivity because reduced leaf area results in decreased chlorophyll content and less photosynthesis. Therefore, leaf area is a good proxy measure for evaluating plant growth and has been used to monitor and estimate crop yield and health [41]. Various other leaf morphological traits exist in plant phenotyping. These include leaf number and inclination angle for monitoring plant growth and development [29, 40], and leaf thickness for measuring abiotic stressors like water scarcity [2]. Additionally, physiological traits of leaves exist that indicate various factors of plant status like stomatal conductance for the detection of biotic stressors like pathogens [14], leaf water content for drought stress [37].

Although leaves play an important role in evaluating plant growth, health and resilience to stress, these evaluations cannot be scaled to the whole plant [40]. Rather, other parts of the plant should be phenotyped to understand the processes governing plant status comprehensively. The literature is rife with such examples that study traits of roots [6], stems [9], seeds [39], and fruit [1].

2 Image Processing And Plant Phenotyping

Traditional plant phenotyping methods are destructive, prone to human error, time-consuming, and challenging to perform from start to end of a plant's ontogenesis [4, 40]. Recent advances in image sensing, processing, and data management have produced high-throughput image-based plant phenotyping systems, which alleviate the issues with traditional manual methods. These high throughput systems are amenable to automation and can analyse a large sample of data precisely in a very short period. Reducing the need for potentially erroneous, time-consuming, expensive human labour [8]. Modern high-throughput image-based plant phenotyping systems are examples of cyberinfrastructure (CI) systems, which are research environments that support data acquisition, storage, management and computing and processing services distributed over some network [25]. For image-based plant phenotyping, this means that image acquisition and processing techniques are vital.

2.1 Image Acquistion In Plant Phenotyping Systems

Fill this out

2.2 Image Processing In Plant Phenotyping Systems

What is the purpose of this section?

- Discuss image processing in the context of plant phenotyping
- Discuss image acquistion techniques
- Discuss image processing techniques used in plant phenotyping

Foundations of image acquistion in plant phenotyping

- measure a phenotype quantitatively through the interaction between light and plants such as reflected photons, absorbed photons, or transmitted photons. Each component of plant cells and tissues has wavelength-specific absorbance, reflectance, and transmittance properties [20]
- For example, chlorophyll absorbs photons primarily in the blue and red spectral region of visible light [20]
- Imaging at different wavelengths is used for different aspects of plant phenotyping [20].
- Imaging techniques are very helpful for detecting these properties, especially for those that cannot be seen by the naked eye [20]

Image acquistion methodologies in plant phenotyping

Thermal imaging sensors

- Conceptual basis of this technique [28]
- Have been used to measure plant water status.
- Used in Irrigation scheduling.
- By inferring transpiration rate of leaves which is governed by stomoatal conductance.
- Stomotal conductance can indicate drought, pathogens, and also soil salinity.
- Have also been used for yield predictions [14]

Hyperspectral sensors

- Based on the principles of spectroscopy. Can detric biochemical, morphological and physilogical traits of plants. Has been used to detect drought stress. Also used to detect water stress, and pathogens. Different reflectance can tell us about different things. Certain band tells us about water stress, some can tell us about clholrphyll content. [32].
- Conceptual basis and applications [31]
- Crop yield
- Biotic and abiotic factors

Tomography

- three-dimensional visualization of plant internal structures, root systems, and physiological processes
- Based on principles of xray cts, moving sensor takes planar cross sections to 3d visialize plant
- Drought stress [35]
- Yield Estimations
- has references for usages [27]

Flouresnce sensors

- Conceptual basis of this technique [33]
- Predict metabolism using photosynthesis which tells us about growth status

- Any stress or disturbance can indirectly or directly affect plant metabolism, so chlorphyll content canbe us used as a measure to detect these things like pathogens, light shortage, nitrogen deficiency, water stress etc [21]
- In optimal conditions, very little heat is emitted in photosynthesis so in certain near infrared emission should be low
- Acitinic light involved here?

RGB Camera

- Mimic human vision
- measure plant shoot biomass (yield potential) [20]
- leaf senance (tissue resistance to salt)

What is the purpose of this section?

- Discuss how image processing and plant phenotyping are applied in low-cost automated greenhouses
- What methods, limitations and advances have there been.
- What does a typical image preprocessing pipeline look like?

Basis for future in low-cost imaging analysis
The basis for the points below is from [25]

- Existing phenotyping systems are comercially developed and under patent, so are not modifable or readily available [18]
- RGB, and hyperspectral sensors most commonly used as they are affordable
- Data collection and analysis tools are not generalizable or standardised, so many custom flows are made
- Indicating the need for standardised, extendable image analysis pipelines using readily avalble open-source softwatre and hardware
- Recenet advances show the use of cheap hardware like raspberry pis, and afforadbale RGB sensors [34]
- However these sensors dont have a lot of compute power, so analysis and preprocessing is done remotely
- Cheap sensors but are chaper and faster
- Cloud archtiectures exist, but with amount of data generated this cost can be prohibitive in terms of processing cost [5]. Cyberinfrastructure expensive ref (95,96)
- Moreover, existing systems are not designed for multiple genotypes so thier applicability is limited, suggesting need for more generalizable image analysis
- Deeplearning and machine learning can be used to automate image analysis, and reduce costs
- Availablity of large labelled data-is bottleneck
- To use on cheap hardware you also need to optimized
- Therefore low complexity effiecinet algorithms for automated image preprocessing and image analysis is needed
- Using multimodal segmenation is also another angle

- ref[36], one of the advantages of using ML approaches in plant phenotyping is their ability to search large datasets and discover patterns by simultaneously looking at a combination of features (compared to analyzing each feature separately). This was previously a challenge because of the high dimensionality of plant images and their large quantity, making them difficult to analyze through traditional processing methods (ref 37)

- machine learning achived using supervised learning, so we arent able to detect novel or unexpected traits using unsupervised learning (ref 40)

- traditional methods in image processing not geeralizable, in pathogen detection and require strict conditions and custom workflows to work [4]

- Current research Focus on image sensing, rather on computation which is new bottleneck. Imaging technlogy has advanced, but there is lack of robust analysis and computation tools. [5]

- Sensors generate huge amount of data, therefore algorithms need to be designed to deal with Terabytes of date. Furthermore, low cost computing units have limited bandwidth, battery and sensor capablities indicating that algorithms need to take this in to account. Low complexity and efficeint algorithms that can run on resource constrained tech is needed [5]

- Mobile applications can be uesd and run processing without need for extranl network, cyberinfrasturcture. Howevere there are not many applications available for this [24]. Image analysis tools are often too expensive to run on these devices due to the constrained resources, however they are much cheaper and therefore can be exploited. In particular low complexict effeicent systems, which generalize to various genotypes, and phenotypcal mesauremtns are needed. In particular the development of effiecnet low resouce intensive machine modelling and deep learning techniques can be made.

- preprocessing technique choice is also needed, for example image segmentation methods can be costly, and a suitable method depends on various factors. Certain methods have cheaper cost, but less accuracy [24]. In developing low-cost image anaysis and preprocessing tools for phenotyping it is neccesary to weight the cost savings versus accuract and precision. But besoke image processing techniques dont lend well to generalziabality and extendability.

- cost-effective solutions require carefully choosing sensors and date processing techniqyes to be most cost-effective [40]

- automated systems not built for general solutions but for specifc species of plants [11]

Current applications

- Often combined with machine learning algorithms
- Cheap Rasberry PI with RGB camera [34]
- Autonomous robots with sensors and cameras [30]
- use off the shelf commerical software like microsoft xbox360 kinect [19]

What affects a low-cost automated greenhouse cost
Points from this paper [16]

- Image preprocessing pipeline
- open source software
- hardware
- amount of labour and time needed for analysis
- enery cost of technology
- amount of data generated, where is it stored, how frequently? [10]

Challenges of low-cost automated greenhouses

- Many phenotyping systems exisiting are comercially developed under patent so hardware and software can not be modified. [18]
- Current applications use moving conveyor belt (benchtop) [18]
- or convery belt [18]
- or moving robotic arm to move sensors around to get high-throughput measurements.
- Robot solutions not cheap for farmers [30]
- large amounts of data need to be colleceted, and processed. Exisitng phenotyping platforms exist, but they are expensive and have restricted running environments.
- Many systems developed for labratory systems, but not extended for real applications. Standarides off the shelf equipment, open-source software allow modular and extensible plus affordable solutions [22]
- Custom data collection schemes need to be made very little standards
- lack of robust methods to analyse plant images
- low cost moving track systems limited in phenotypical, genotype and management types are supported [38] suggesting need for low-cost systems that can support a wider variety of genetypes and phenotypical traits.

Fill out the conclusion section.

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